Zihan (Tomson) Li

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Education

Washington University in St. Louis — St. Louis, MO
Doctor of Philosophy in Computer Science — *Aug. 2023 – Present (Expected May 2028)*

Washington University in St. Louis — St. Louis, MO
Master of Science in Cybersecurity Engineering — *Aug. 2022 – May 2023*

Washington University in St. Louis — St. Louis, MO
Bachelor of Science in Computer Engineering — *Aug. 2020 – Dec 2022*

DePauw University — Greencastle, IN
Bachelor of Arts — *Aug. 2017 – May 2020*

Experience

Graduate Research Assistant — May 2022 – Present
Washington University in St. Louis — *St. Louis, MO*

- Research focuses on **system security and cyber-physical systems (CPS).**
- Analyzed IoT firmware updates, examining 150 images from 33 device families. Discovered **zero-day and n-day** vulnerabilities, leading to **25 CVE IDs and one PSV ID** through responsible disclosure.
- Developed experimental Linux scheduler enforcer for **timing violation detection and mitigation** with **an average performance overhead of only around 2.8%**.
- Optimization on communication cost of Federated Learning. Reducing communication cost by **30%** while maintaining training accuracy of **95%**.
- Discovery of timing impact on security protection mechanisms for CPS. Developed an optimization framework that **mitigates attacks while maintaining schedulability** for **real-time CPS**.

C++ Backend Development Intern — July 2020 – Sept. 2020
Youme.im — *Shenzhen, China*

- Developed an **end-to-end encryption module** for audio and video real-time communication
- Integrated the encryption module into the cross-platform compilation build workflow
- Utilizing optimization techniques to ensure **low-performance overhead** for encryption and decryption.

Publications

Conference Papers

Balancing Security and Schedulability: WCET Evaluation and Security Optimization in CPS
2026 IEEE 32nd Real-Time and Embedded Technology and Applications Symposium (RTAS)
Zihan Li, Marion Sudvarg, Ching-Hsiang Chan, Ryan Burrow, Nathan Burow, Cailani Lemieux-Mack, Sanjoy Baruah, Ning Zhang, Bryan C. Ward

Sensing and Stopping Interfering Secondary Users: Validation of an Efficient Spectrum Sharing System
2026 IEEE International Symposium on Spectrum Innovation (DySPAN)
Meles G Weldegebriel, **Zihan Li**, Dustin Maas, Gregory Hellbourg, Ning Zhang, Neal Patwari

Resilient Federated Learning on Embedded Devices with Constrained Network Connectivity
2025 62nd ACM/IEEE Design Automation Conference (DAC)
Zihan Li, Han Liu, Ao Li, Ching-Hsiang Chan, Yevgeniy Vorobeychik, William Yeoh, Wenjing Lou, Ning Zhang

Tintin: A Unified Hardware Performance Profiling Infrastructure to Uncover and Manage Uncertainty
19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25)
Ao Li, Marion Sudvarg, **Zihan Li**, Sanjoy Baruah, Chris Gill, Ning Zhang

Your Firmware Has Arrived: A Study of Firmware Update Vulnerabilities
33rd USENIX Security Symposium (USENIX Security 24)
Yuhao Wu, Jinwen Wang, Yujie Wang, Shixuan Zhai, **Zihan Li**, Yi He, Kun Sun, Qi Li, Ning Zhang

Conference / Workshop Short Papers

Spectrogram-Based Interference Source Attribution at OVRO: An Over-the-Air Validation Using Pseudonymetry
To Appear in 2026 IEEE RFID NRDZ Workshop
Meles G Weldegebriel, **Zihan Li**, Gregory Hellbourg, Ning Zhang, Neal Patwari

Tintin: PMU Scheduling to Minimize Uncertainty
OSPERT Workshop at ECRTS 2025
Marion Sudvarg, Ao Li, **Zihan Li**, Sanjoy Baruah, Chris Gill, Ning Zhang

Work-in-Progress: Measuring Security Protection in Real-time Embedded Firmware
2022 IEEE Real-Time Systems Symposium (RTSS)
Yuhao Wu, Yujie Wang, Shixuan Zhai, **Zihan Li**, Ao Li, Jinwen Wang, Ning Zhang

Projects

Open Platform for Cyber Physical System Research (OPCPS) April 2025 – Present

- Design and implement an open-source platform specifically tailored for CPS security research testing
- Offers modular components designed for trivial replacement with alternative implementations.
- Security instrumentations for various real-world CPS platforms
- Performance and timing profiling to understand the real-time impact of CPS under different security instrumentations.

EMILY: Electro Magnetic Interference Ledger & registrY | *EMI, Database* Nov 2025 – Present

- A database to provide interference insights for radio astronomy.
- Design database structure to improve the performance and maintainability of EMILY.
- Implement backend for high performance and secure access to the database based on Springboot framework.
- Design and implement a user-friendly front-end for astronomers to use the system with ease.
- Real-world impact and public available at emilyknows.info

Federated Learning Optimization | *Machine Learning, Federated Learning, Python* Aug 2024 – Dec 2024

- Designed adaptive ML protocols for distributed predictive modeling, using **PyTorch, pandas, and NumPy**. Cut communication costs by **30%** while preserving **95%** training accuracy.
- Conduct an empirical study on communication cost for federated learning.
- Designed and optimized data pipelines for processing large-scale distributed datasets. Adaptive gradient compression rate can reach up to **210x**
- Real-world FL simulation demonstrates the feasibility of the proposed approach.
- Accepted by *2025 62th ACM/IEEE Design Automation Conference (DAC)*

Honor Awards

2022 Dean's Select PhD Fellowship | *Washington University in St. Louis*

Nominated for the 2022 Dean's Select PhD Fellowship at Washington University in St. Louis.

Dean's List | *DePauw University*

Recognized on the Dean's List for 2017 and 2020

Services

Reviwer

- IEEE Transactions on Information Forensics and Security
- IEEE/ACM Transactions on Networking
- ACM Transactions on Cyber-Physical Systems
- ISOC Symposium on Vehicle Security and Privacy (VehicleSec '24)
- International Conference on Computer Communications and Networks (ICCCN)

Artifact Evaluation Program Committee

- ACM Conference on Computer and Communications Security (CCS)

Teaching

CSE 5619: Recent Advances in Computer Security and Privacy Spring 2026

- AI (Assistant in Instruction) and class coordinator

Technical Skills

Languages: C/C++, Python, Java, JavaScript, HTML/CSS, VHDL, Assembly, SQL, PHP
Frameworks: React, Node.js, ROS, ROS2
Developer Tools: Git, Cmake, Docker, VS Code, Visual Studio, Eclipse, Wireshark, Xcode, Ghidra, Database Management Systems, Excel, Gazebo
Libraries: pandas, NumPy, Matplotlib, Tkinter, Pytorch
OS: Linux Kernel Programming, Kernel Scheduler, Kernel Network Stack